



Review

- In the last lesson, you learned that the optimal Bayesian recursion is to:
 - Step 1: Compute the pdf for predicting x_k given all *past* observations:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}.$$

- Step 2: Update the pdf for estimating x_k given *all* observations via:

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- If we knew how to compute these pdfs, then we could find any desired estimator; e.g., mean, median, mode, whatever.
- In this lesson, we begin to develop a framework for approximating these pdfs.



Numeric integration

- The prediction step requires evaluating the integral:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}.$$

- Normalization of $f(x_k | \mathbb{Z}_k)$ via its denominator requires evaluating the integral:

$$f(z_k | \mathbb{Z}_{k-1}) = \int_{R_{x_k}} f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1}) dx_k.$$

- If state dimension n is large (e.g., $n \geq 3$), numeric evaluation of these multi-dimensional integrals may be impractical, even with modern CPUs.
- Computation needed for a given precision scales exponentially with n .
 - The particle filter will explore some clever ways to avoid this issue.
 - But, first, we look at brute-force numeric integration for low-order problems.



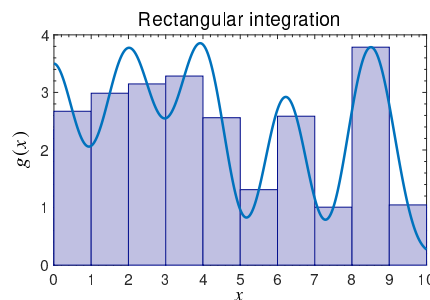
Approximating using the rectangular rule

- Suppose that we wish to approximate the definite integral:

$$\text{integral} = \int_{x_{\min}}^{x_{\max}} g(x) dx.$$

- Let N_r be the number of regions each having fixed width $w = (x_{\max} - x_{\min})/N_r$.
- The rectangular rule approximates $g(x)$ as piecewise constant.
- Then, the integral can be approximated as:

$$\text{integral} \approx \sum_{i=0}^{N_r-1} w \cdot g(x_{\min} + (i + 1/2)w).$$





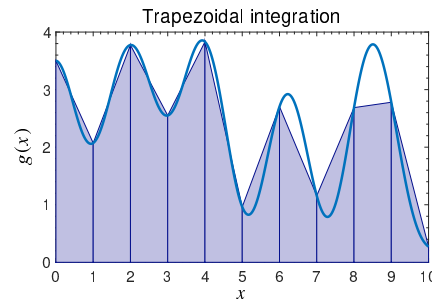
Approximating using the trapezoidal rule

- Again, we wish to approximate the definite integral:

$$\text{integral} = \int_{x_{\min}}^{x_{\max}} g(x) dx.$$

- The trapezoidal rule approximates $g(x)$ as piecewise linear.
- The integral approximation is:

$$\text{integral} \approx w \left[\frac{g(x_{\min}) + g(x_{\max})}{2} + \sum_{i=1}^{N_r-1} g(x_{\min} + iw) \right].$$



Calculations with rectangular rule

- For the illustrations on earlier slides, the table gives x , $g(x)$.
- We wish to approximate the actual integral of $g(x)$ over the span of 0 to 10 (which is 24.4030) using the rectangular rule.
- There are $N_r = 10$ regions and the width of each region is:

$$w = \frac{x_{\max} - x_{\min}}{N_r} = \frac{10 - 0}{10} = 1.$$

- We approximate the integral using the rectangular rule:

$$\begin{aligned} \text{integral} &\approx \sum_{i=0}^{N_r-1} w \cdot g(x_{\min} + (i + 1/2)w) \\ &= 1 \cdot (2.6729 + 2.9861 + \dots + 1.0457) \\ &= 24.3866. \end{aligned}$$

x	$g(x)$
0.5	2.6729
1.5	2.9861
2.5	3.1475
3.5	3.2844
4.5	2.5601
5.5	1.3104
6.5	2.5869
7.5	1.0066
8.5	3.7860
9.5	1.0457



Calculations with trapezoid rule

- For the illustrations on earlier slides, the table gives x , $g(x)$.
- We wish to approximate the actual integral of $g(x)$ over the span of 0 to 10 (which is 24.4030) using the trapezoid rule.
- Again, $N_r = 10$ and $w = 1$ and we now approximate the integral using the trapezoid rule:

$$\begin{aligned} \text{integral} &\approx w \left[\frac{g(x_{\min}) + g(x_{\max})}{2} + \sum_{i=1}^{N_r-1} g(x_{\min} + iw) \right] \\ &= 1 \left[\frac{3.5015 + 0.2720}{2} + (2.0722 + \dots + 2.7806) \right] \\ &= 24.4064. \end{aligned}$$

- In this case, the trapezoid rule is more accurate than the rectangular rule (generally true when w becomes small).

x	$g(x)$
0.0	3.5015
1.0	2.0722
2.0	3.7745
3.0	2.5508
4.0	3.8308
5.0	0.9628
6.0	2.7045
7.0	1.1563
8.0	2.6872
9.0	2.7806
10.0	0.2720



Summary

- Implementing the Bayesian-inference solution requires computing or approximating a number of (possibly multi-dimensional) integrals.
- This week, we will take the approach of approximating the integrals in a direct way.
 - We can use either rectangular or trapezoidal approximations.
 - When the region width w is small, trapezoidal approximations are more accurate, so we will proceed with those as we move forward.